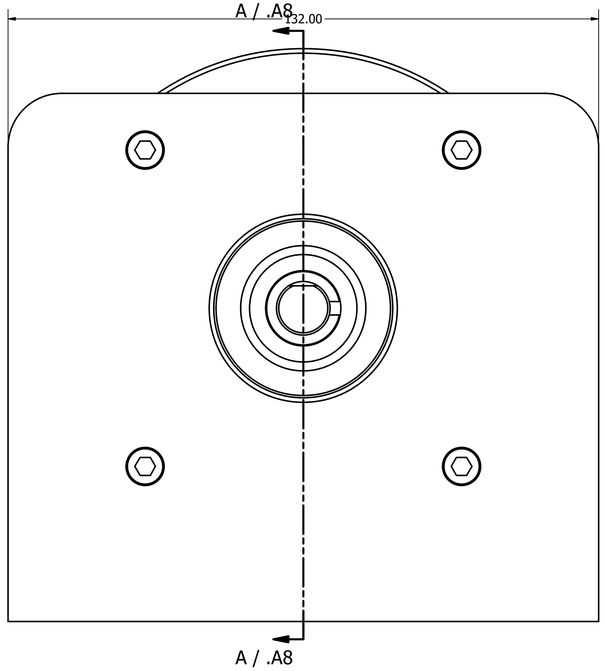
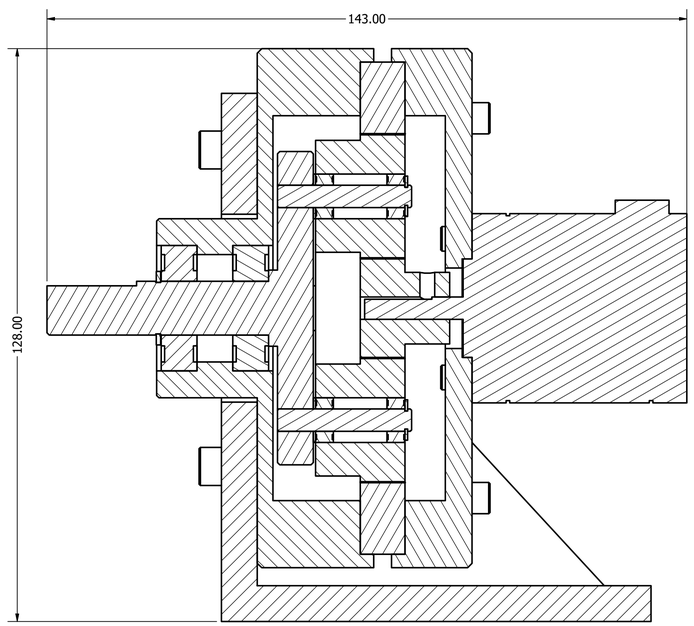
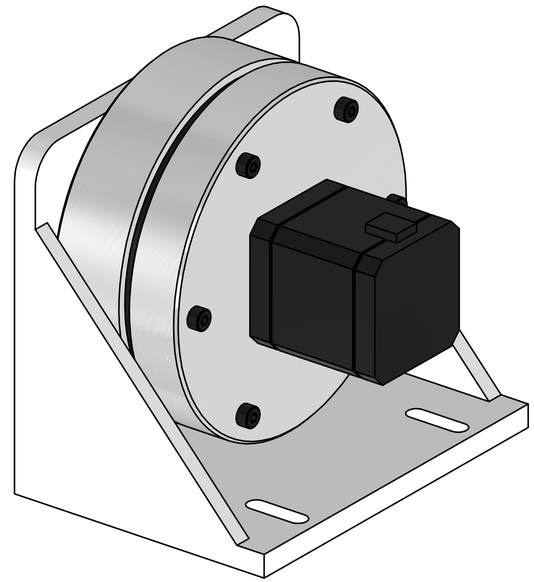




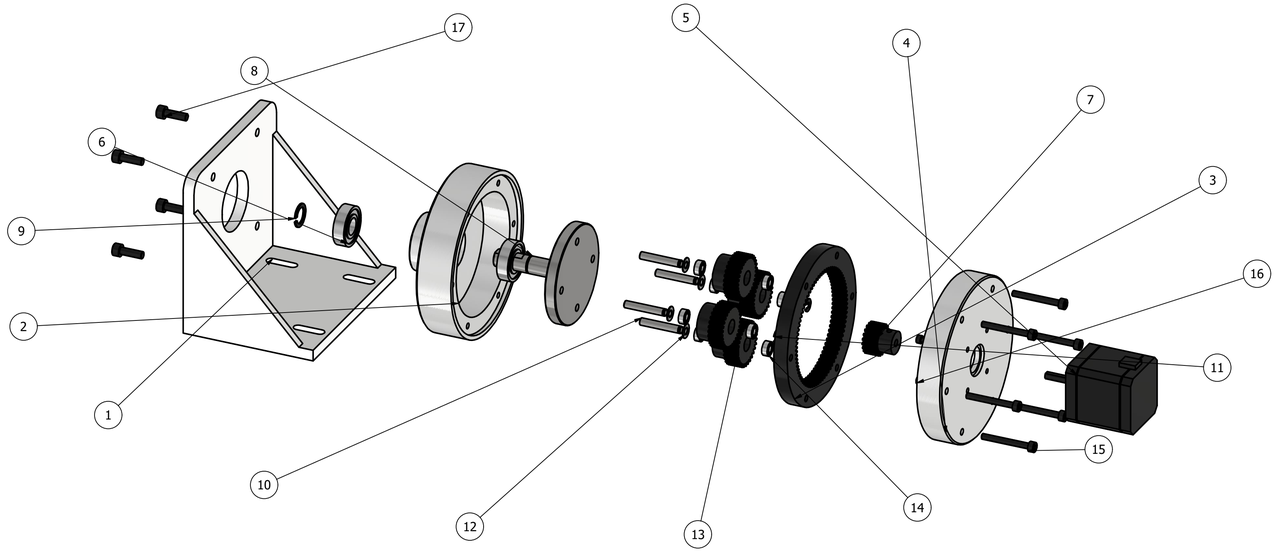
SECTION A-A
SCALE 1.50 : 1 / .C7



PLANETARYGEARBOX - SINGLE-STAGE PLANETARY REDUCER 5:1 FOR A NEMA 17 STEPPER (DESIGNED, NOT BUILT)
GEARS KHK m1 20 DEG S45C: SUN SS1-20A (z20, ON THE 5 MM MOTOR D-SHAFT), 4 PLANETS SS1-30 (z30), RING SI1-80 (z80, FIXED)
RATIO 1 + 80/20 = 5.0; CENTRE DISTANCE 25; PLANETS ON 2 X MR105ZZ EACH, 5 MM PINS PRESSED IN THE 1045 CARRIER
OUTPUT 12 MM SHAFT ON 2 X 6001-2RSH; PERMISSIBLE OVERHUNG LOAD 200 N AT 12 MM FROM THE NOSE (CALC.)
RATING (CALC.): 5.1 NM CONTINUOUS (KHK SUN SURFACE DURABILITY), MOTOR-LIMITED 2.83 NM;
DESIGN POINT 60 RPM OUT, 1.89 NM, EFFICIENCY 95 % (CALC.)
BACKLASH AT THE OUTPUT 12-28 ARCMIN FROM KHK TOLERANCES (CALC.)
GREASE: NLGI 1 EP, FILL THE GEAR CHAMBER ABOUT 1/3. LOCTITE 641 ON THE MR105ZZ OUTER RINGS.
ASSEMBLY ORDER: MOTOR + ADAPTER (4 X M3) > SUN ON THE D-FLAT (M4 SET SCREW) > CARRIER WITH PLANETS INTO THE RING > HOUSING + BEARINGS > 6 X M4 X 35 THROUGH ADAPTER AND RING. ASSEMBLY PARAMETER: Input (MOTOR ANGLE).

DRAWN Jay Patel	10/28/2026	
CHECKED		TITLE: PlanetaryGearbox - general assembly
QA		
MFG		
APPROVED		
		SIZE D · DWG NO PG-DWG-001 · REV
		SCALE 1.50 : 1 · SHEET 1 OF 2

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PG-105	Bench mount bracket, PETG printed (upright face down on the bed): 42 nose hole, 4 x 5.5 on 100 PCD, 4 x M6 slots, 2 gussets
2	1	PG-102	Output housing, 6061-T6: 110 H7 ring recess, 6 x M4 clamp threads, 2 x 6001 seats 28 J7 with a 24 shoulder, 4 x M5 mounting face on 100 PCD
3	1	PG-213	Ring gear KHK SI1-80: internal m1 z80, S45C black oxide, 110 OD x 10; MODIFIED: 6 x 4.5 thru on 98 PCD for the clamp bolts
4	1	PG-101	Motor adapter, 6061-T6: NEMA 17 face (22 H7 pilot, 4 x M3 on 31), 110 H7 recess 3 deep for the ring, 6 x 4.5 on 98 PCD
5	1	PG-201	Stepper motor StepperOnline 17HS19-2004S1: NEMA 17, 0.59 Nm holding, 2.0 A, 1.6 ohm, 3.0 mH, 5 mm D-shaft x 24 (datasheet A0842); radial load 21 N max at 20 mm
6	2	PG-222	Ball bearing SKF 6001-2RSH 12 x 28 x 8, sealed, C 5.4 kN, C0 2.36 kN; output shaft (2)
7	1	PG-211	Sun gear KHK SS1-20A: spur m1 z20, 20 deg, S45C black oxide, 10 face, bore 5 H8 + M4 set screw onto the motor D-flat
8	1	PG-103	Planet carrier with integral 12 mm output shaft, 1045 CD: 70 x 8 plate, 4 x 5 P7 pin bores (press) on 50 PCD, 6001 seats 12 k5, DIN 471 groove, 1 mm D-flat
9	1	PG-231	Retaining ring DIN 471 12 x 1 (groove 11.5 x 1.1), McMaster 98541A119
10	4	PG-104	Planet pin 5 h6 x 30, hardened ground shaft (Misumi SFJ/PSFJ 5 mm class, cut to length), DIN 6799 groove; press fit into the carrier
11	4	PG-232	Retaining washer DIN 6799 RS 4 (E-clip, groove 4 x 0.64) on the planet pin
12	4	PG-233	Shim ring DIN 988 5 x 10 x 0.5 between planet bearing and carrier
13	4	PG-212	Planet gear KHK SS1-30: spur m1 z30, 20 deg, S45C black oxide, 10 face, plain bore 10 (2 x MR105ZZ, Loctite 641)
14	8	PG-221	Ball bearing NSK MR105ZZ 5 x 10 x 4, shielded, Cr 475 N, C0r 168 N (nsk.com); 2 per planet
15	6	PG-241	Socket head cap screw ISO 4762 M4 x 35, 12.9 (adapter + ring into the housing, 6)
16	4	PG-242	Socket head cap screw ISO 4762 M3 x 8, 12.9 (adapter to motor, 4)
17	4	PG-243	Socket head cap screw ISO 4762 M5 x 16, 12.9 + washer (bracket to housing, 4)



DRAWN Jay Patel	10/28/2026		
CHECKED			
QA		TITLE	
MFG		PlanetaryGearbox - general assembly	
APPROVED		SIZE D	DWG NO PG-DWG-001 REV
		SCALE 1 / 2	SHEET 2 OF 2